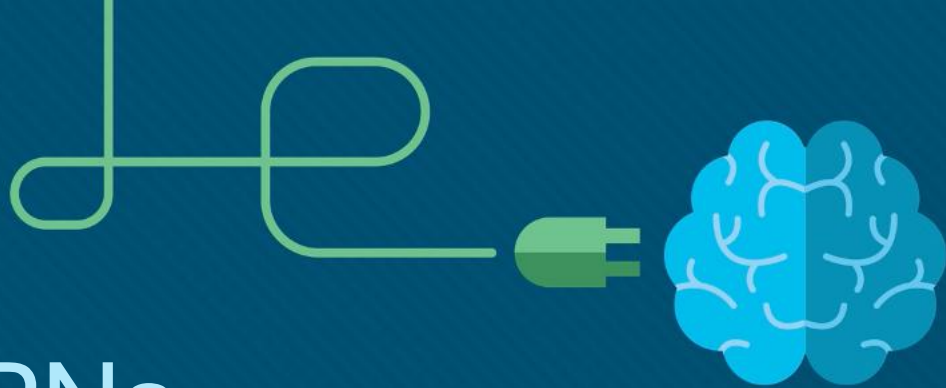




Module 18: VPNs

Networking Security v1.0
(NETSEC)



Module Objectives

Module Title: VPNs

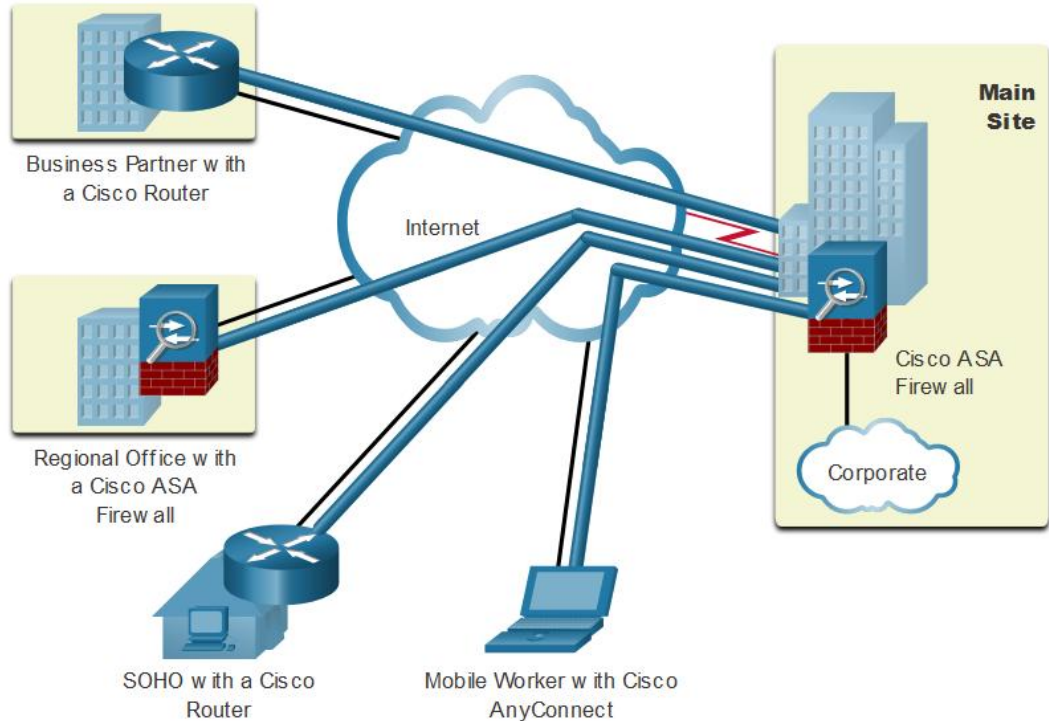
Module Objective: Explain the purpose of VPNs.

Topic Title	Topic Objective
VPN Overview	Describe VPNs and their benefits.
VPN Topologies	Compare remote-access and site-to-site VPNs.
Introducing IPsec	Describe the IPsec protocol and its basic functions.
IPsec Protocols	Compare AH and ESP protocols.
Internet Key Exchange	Describe the IKE protocol.

18.1 VPN Overview

Virtual Private Networks

To secure network traffic between sites and users, organizations use virtual private networks (VPNs) to create end-to-end private network connections. A VPN is virtual in that it carries information within a private network, but that information is transported over a public network. A VPN is private in that the traffic is encrypted to keep the data confidential while it is transported across the public network. The figure shows a collection of various types of VPNs managed by an enterprise's main site. The tunnel enables remote sites and users to access main site's network resources securely.



VPN Benefits

Modern VPNs now support encryption features, such as Internet Protocol Security (IPsec) and Secure Sockets Layer (SSL) VPNs to secure network traffic between sites.

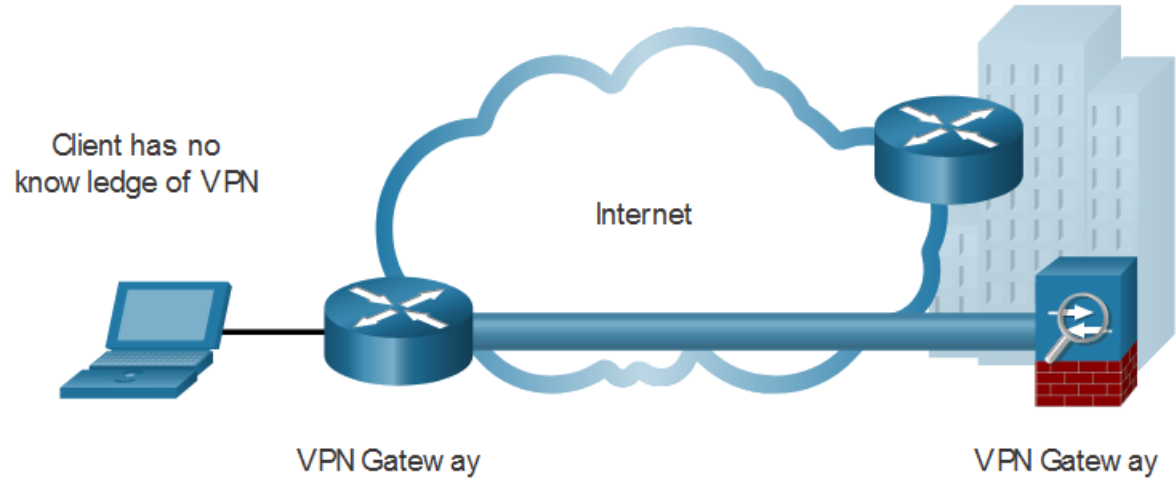
Major benefits of VPNs are shown in the table.

Benefit	Description
Cost Savings	With the advent of cost-effective, high-bandwidth technologies, organizations can use VPNs to reduce their connectivity costs while simultaneously increasing remote connection bandwidth.
Security	VPNs provide the highest level of security available, by using advanced encryption and authentication protocols that protect data from unauthorized access.
Scalability	VPNs allow organizations to use the internet, making it easy to add new users without adding significant infrastructure.
Compatibility	VPNs can be implemented across a wide variety of WAN link options including all the popular broadband technologies. Remote workers can take advantage of these high-speed connections to gain secure access to their corporate networks.

18.2 VPN Topologies

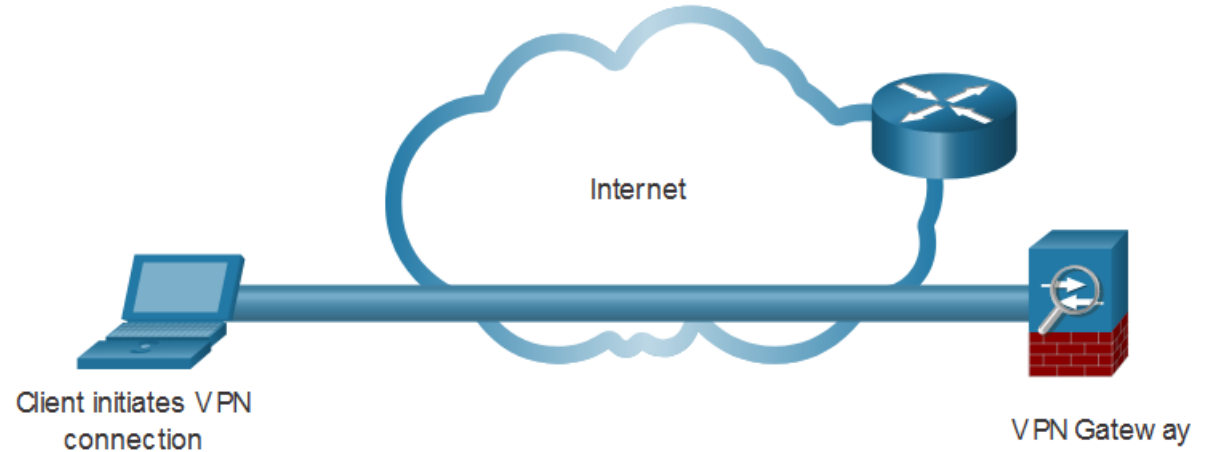
Site-to-Site and Remote-Access VPNs

A site-to-site VPN is created when VPN terminating devices, also called VPN gateways, are preconfigured with information to establish a secure tunnel. VPN traffic is only encrypted between these devices. Internal hosts have no knowledge that a VPN is being used.



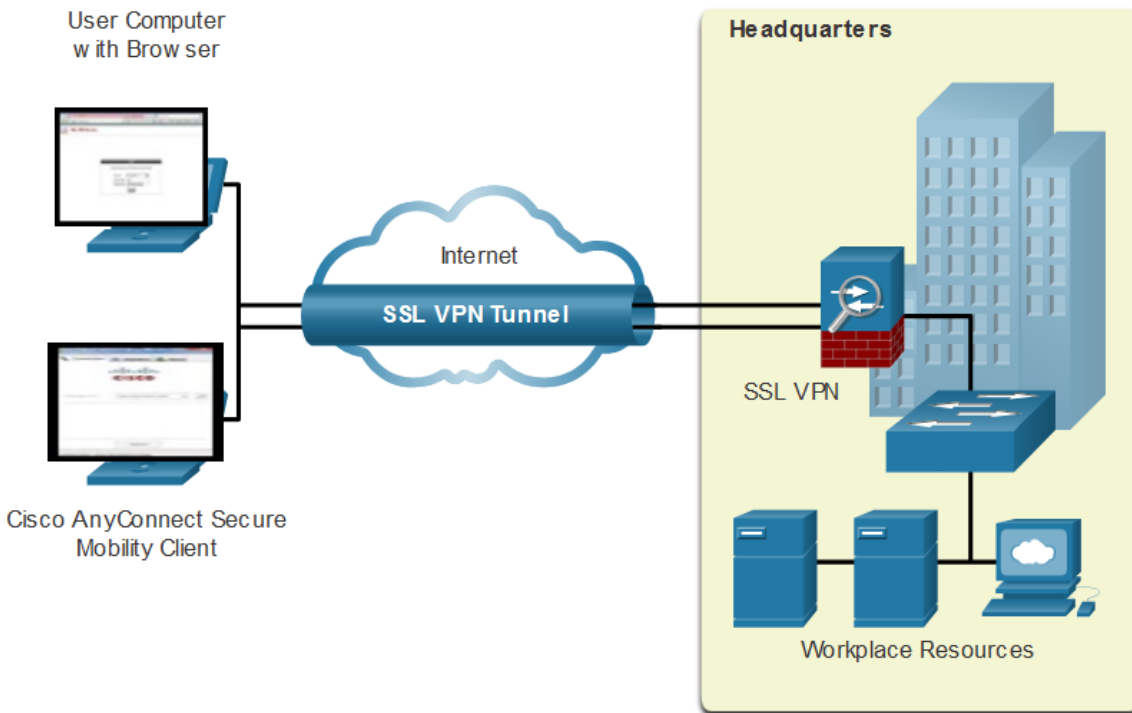
Site-to-Site and Remote-Access VPNs (Cont.)

A remote-access VPN is dynamically created to establish a secure connection between a client and a VPN terminating device. For example, a remote access SSL VPN is used when you check your banking information online.



Remote-Access VPNs

Remote-access VPNs are typically enabled dynamically by the user when required. Remote access VPNs can be created using either IPsec or SSL. A remote user must initiate a remote access VPN connection. The figure displays two ways that a remote user can initiate a remote access VPN connection: clientless VPN and client-based VPN.



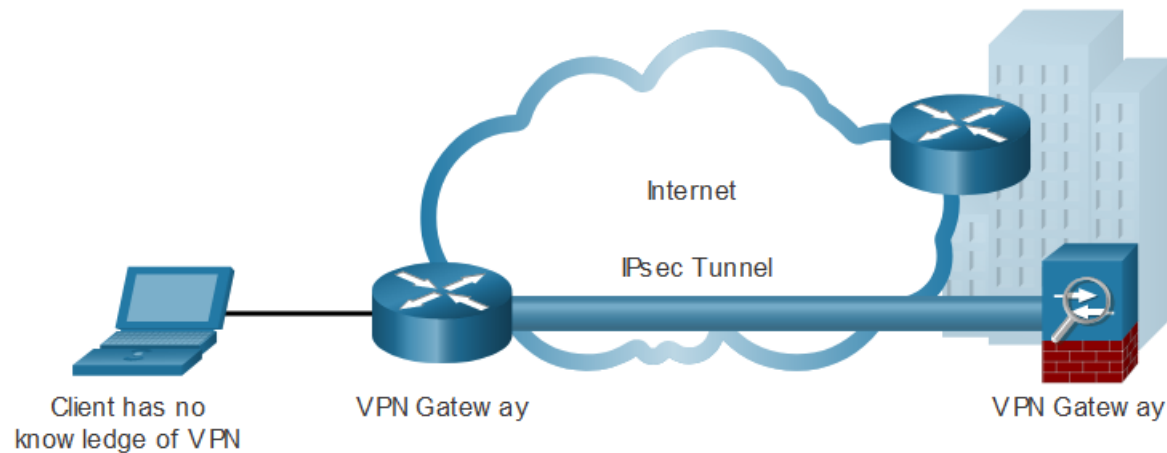
SSL VPNs

SSL uses the public key infrastructure and digital certificates to authenticate peers. Both IPsec and SSL VPN technologies offer access to virtually any network application or resource. However, when security is an issue, IPsec is the superior choice. If support and ease of deployment are the primary issues, consider SSL. The type of VPN method implemented is based on the access requirements of the users and the organization's IT processes. The table compares IPsec and SSL remote access deployments.

Feature	IPsec	SSL
Applications supported	Extensive - All IP-based applications are supported.	Limited - Only web-based applications and file sharing are supported.
Authentication strength	Strong - Uses two-way authentication with shared keys or digital certificates.	Moderate - Using one-way or two-way authentication.
Encryption strength	Strong - Uses key lengths from 56 bits to 256 bits.	Moderate to strong - With key lengths from 40 bits to 256 bits.
Connection complexity	Medium - Because it requires a VPN client pre-installed on a host.	Low - It only requires a web browser on a host.
Connection option	Limited - Only specific devices with specific configurations can connect.	Extensive - Any device with a web browser can connect.

Site-to-Site IPsec VPNs

The VPN gateway encapsulates and encrypts outbound traffic. It then sends the traffic through a VPN tunnel over the internet to a VPN gateway at the target site. Upon receipt, the receiving VPN gateway strips the headers, decrypts the content, and relays the packet toward the target host inside its private network. Site-to-site VPNs are typically created and secured using IP security (IPsec).



18.3 IPsec Overview

Video - IPsec Concepts

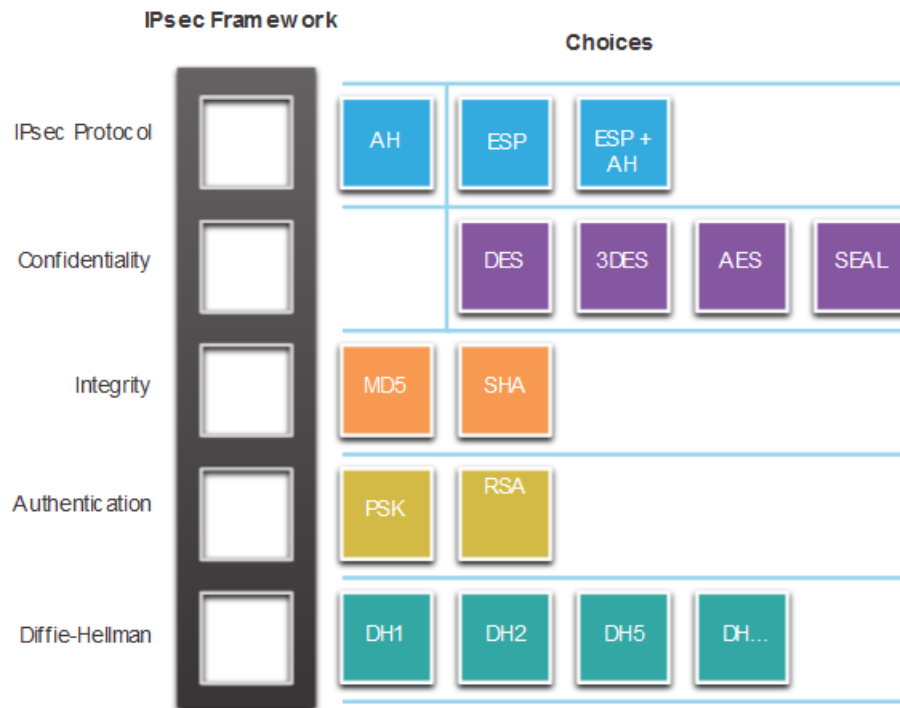
This video will cover the following:

- The purpose of IPsec
- IPsec Protocols

IPsec Technologies

Using the IPsec framework, IPsec provides these essential security functions:

- **Confidentiality** - IPsec uses encryption algorithms to prevent cybercriminals from reading the packet contents.
- **Integrity** - IPsec uses hashing algorithms to ensure that packets have not been altered between source and destination.
- **Origin authentication** - IPsec uses the Internet Key Exchange (IKE) protocol to authenticate source and destination. Methods of authentication including using pre-shared keys (passwords), digital certificates, or RSA certificates.
- **Diffie-Hellman** - Secure key exchange typically using various groups of the DH algorithm.



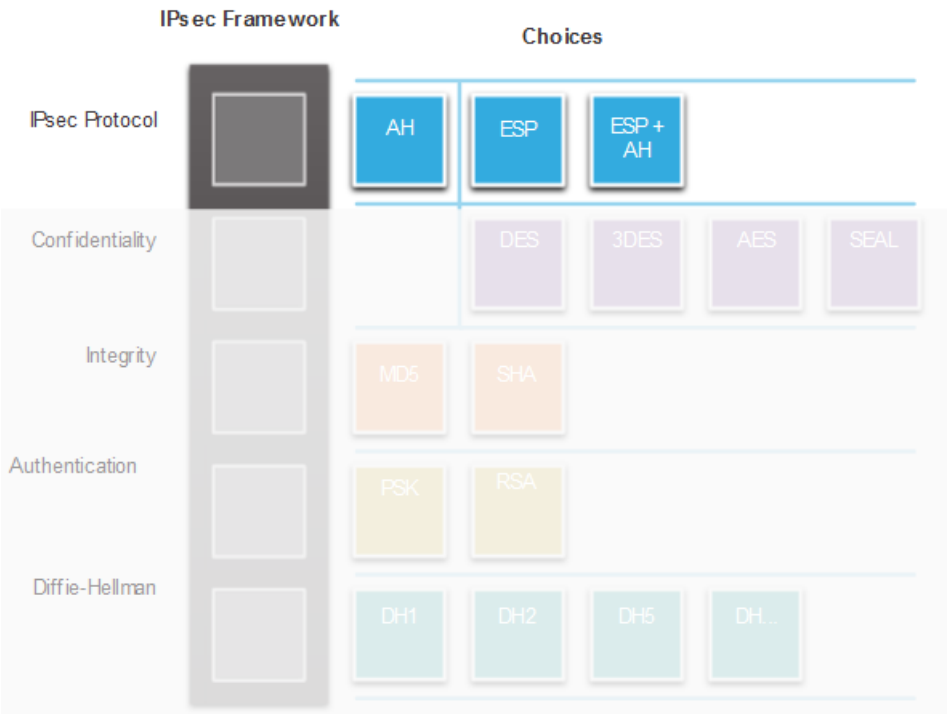
IPsec Technologies (Cont.)

The IPsec security functions are list in the table.

IPsec Function	Description
IPsec Protocol	The choices for IPsec Protocol include Authentication Header (AH) or Encapsulation Security Protocol (ESP). AH authenticates the Layer 3 packet. ESP encrypts the Layer 3 packet. Note: ESP+AH is rarely used as this combination will not successfully traverse a NAT device.
Confidentiality	Encryption ensures confidentiality of the Layer 3 packet. Choices include Data Encryption Standard (DES), Triple DES (3DES), Advanced Encryption Standard (AES), or Software-Optimized Encryption Algorithm (SEAL).
Integrity	Ensures that data arrives unchanged at the destination using a hash algorithm, such as message-digest 5 (MD5) or Secure Hash Algorithm (SHA).
Authentication	IPsec uses Internet Key Exchange (IKE) to authenticate users and devices that can carry out communication independently. IKE uses several types of authentication, including username and password, one-time password, biometrics, pre-shared keys (PSKs), and digital certificates using the Rivest, Shamir, and Adleman (RSA) algorithm.
Diffie-Hellman	IPsec uses the DH algorithm to provide a public key exchange method for two peers to establish a shared secret key. There are several different groups to choose from including DH14, 15, 16 and DH 19, 20, 21 and 24. DH1, 2 and 5 are no longer recommended.

IPsec Protocol Encapsulation

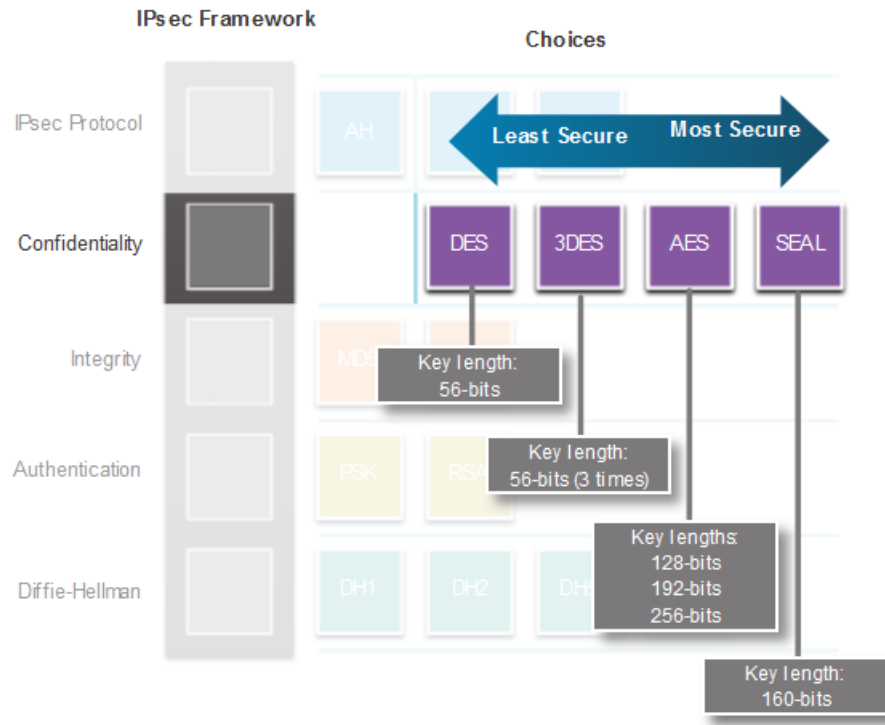
Choosing the IPsec protocol encapsulation is the first building block of the framework. IPsec encapsulates packets using Authentication Header (AH) or Encapsulation Security Protocol (ESP). The choice of AH or ESP establishes which other building blocks are available. AH is appropriate only when confidentiality is not required or permitted. ESP provides both confidentiality and authentication.



Confidentiality

Confidentiality is achieved by encrypting the data. The degree of confidentiality depends on the encryption algorithm and the length of the key used in the encryption algorithm. The shorter the key, the easier it is to break.

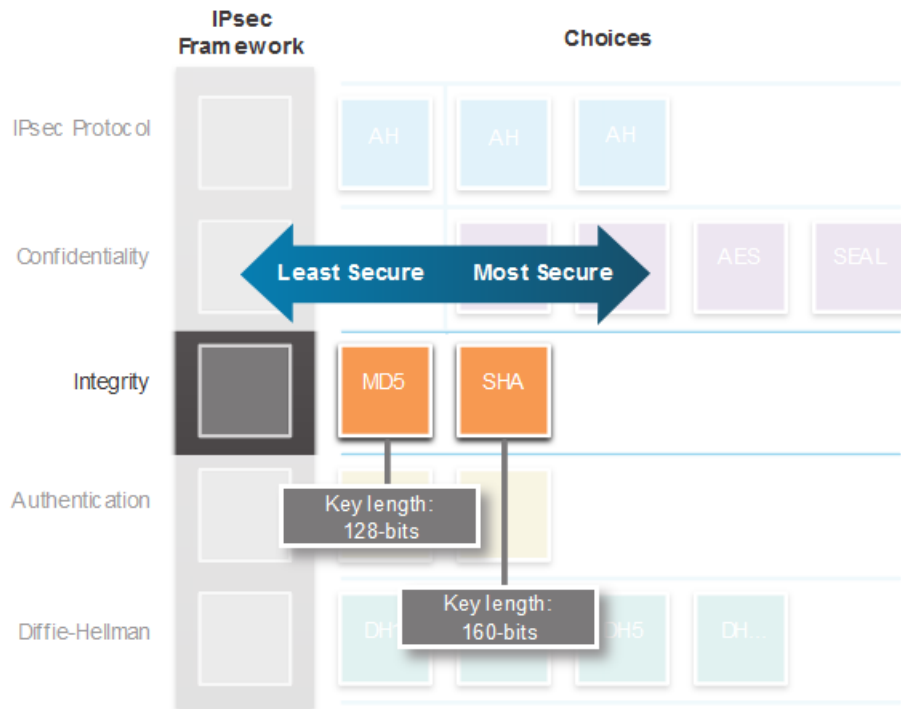
- DES uses a 56-bit key and should be avoided.
- 3DES uses three 56-bit encryption keys per 64-bit block.
- AES offers three different key lengths: 128 bits, 192 bits, and 256 bits.
- SEAL is a stream cipher, encrypting data continuously. It uses a 160-bit key.



Integrity

Data integrity means that the data that is received is exactly the same data that was sent.

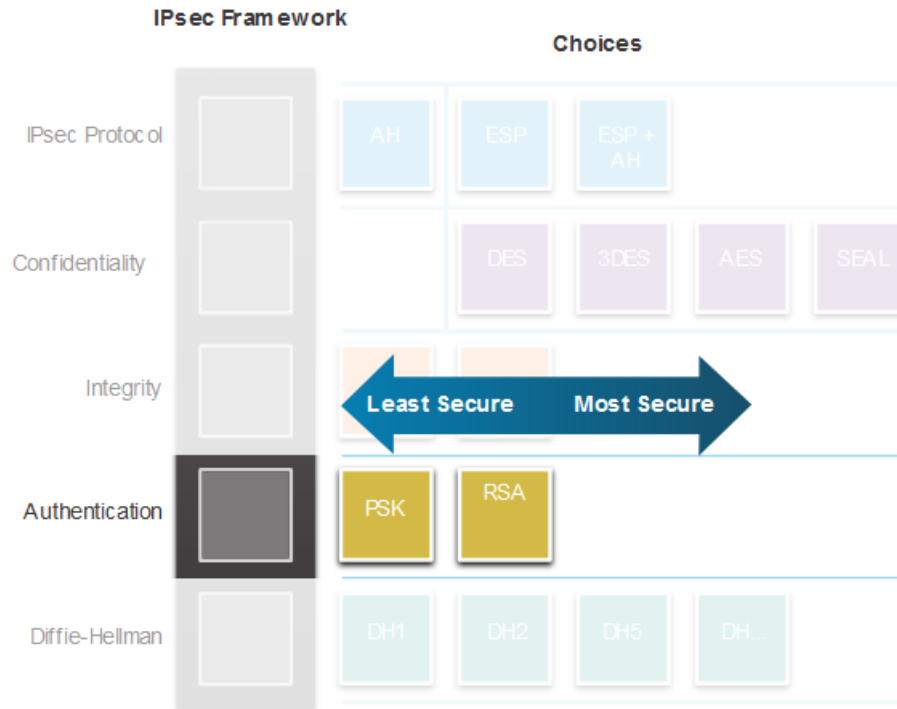
- Message-Digest 5 (MD5) uses a 128-bit shared-secret key. The variable-length message and 128-bit shared-secret key are combined and run through the HMAC-MD5 hash algorithm producing a 128-bit hash.
- The Secure Hash Algorithm (SHA) uses a 160-bit secret key. The variable-length message and 160-bit shared-secret key are combined and run through the HMAC-SHA-1 algorithm producing a 160-bit hash.



Authentication

When conducting business long distance, the device on the other end of the VPN tunnel must be authenticated before the communication path is considered secure.

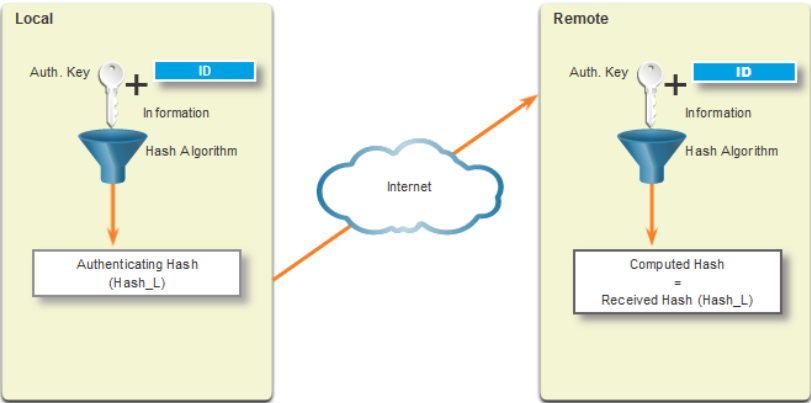
- A pre-shared secret key (PSK) value is entered into each peer manually. The PSK is combined with other information to form the authentication key.
- Rivest, Shamir, and Adleman (RSA) authentication uses digital certificates to authenticate peers.



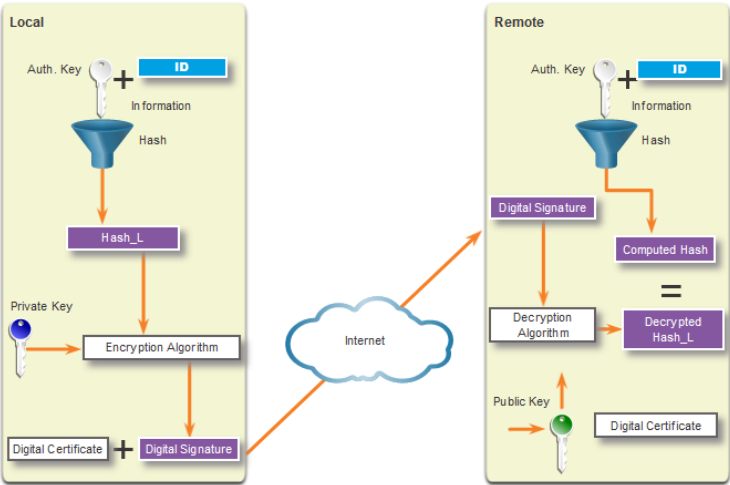
IPsec Overview

Authentication (Cont.)

PSK Authentication

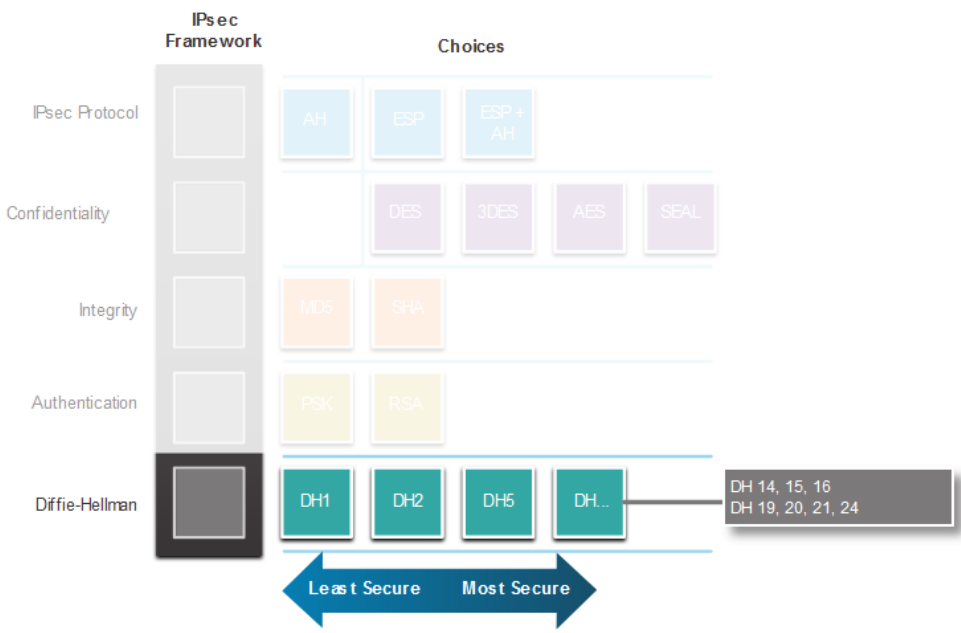


RSA Authentication



Secure Key Exchange with Diffie-Hellman

Encryption algorithms require a symmetric, shared secret key to perform encryption and decryption. How do the encrypting and decrypting devices get the shared secret key? The easiest key exchange method is to use a public key exchange method, such as Diffie-Hellman (DH). DH provides a way for two peers to establish a shared secret key that only they know, even though they are communicating over an insecure channel. Variations of the DH key exchange are specified as DH groups. The DH group you choose must be strong enough, or have enough bits, to protect the IPsec keys during negotiation.



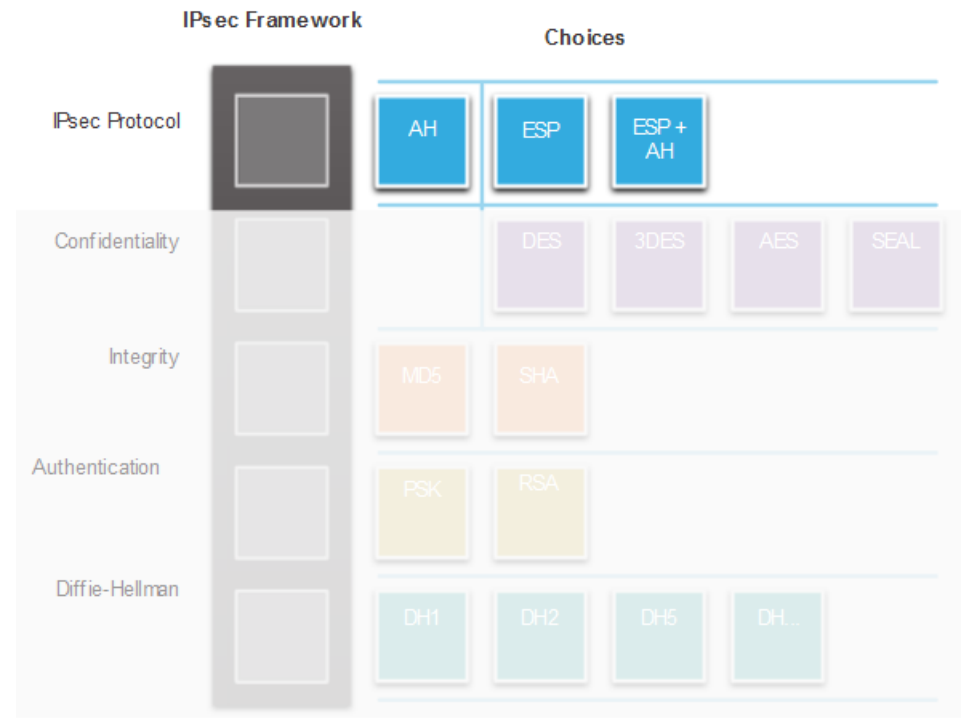
Video - IPsec Transport and Tunnel Modes

This video will explain the process of IPv4 packet with ESP in transport mode and in tunnel mode.

18.4 IPsec Protocols

IPsec Protocol Overview

The two main IPsec protocols are Authentication Header (AH) and Encapsulation Security Protocol (ESP). The IPsec protocol is the first building block of the framework. The choice of AH or ESP establishes which other building blocks are available. AH uses IP protocol 51 and is appropriate only when confidentiality is not required or permitted. It provides data authentication and integrity, but it does not provide data confidentiality (encryption). All text is transported unencrypted. ESP uses IP protocol 50 and provides both confidentiality and authentication. Authentication provides data origin authentication and data integrity.



Authentication Header

The AH function is applied to the entire packet, except for any IP header fields that normally change in transit. Fields that normally change during transit are called mutable fields. For example, the Time to Live (TTL) field is considered mutable because routers modify this field. The AH process occurs in this order:

1. The IP header and data payload are hashed using the shared secret key.
2. The hash builds a new AH header, which is inserted into the original packet.
3. The new packet is transmitted to the IPsec peer router.
4. The peer router hashes the IP header and data payload using the shared secret key, extracts the transmitted hash from the AH header, and compares the two hashes.

The hashes must match exactly. If one bit is changed in the transmitted packet, the hash output on the received packet changes and the AH header will not match. AH supports MD5 and SHA algorithms.

Encapsulation Security Protocol

If ESP is selected as the IPsec protocol, an encryption algorithm must also be selected. The default algorithm for IPsec is 56-bit DES. ESP can also provide integrity and authentication. First, the payload is encrypted. Next, the encrypted payload is sent through a hash algorithm. The hash provides authentication and data integrity for the data payload.

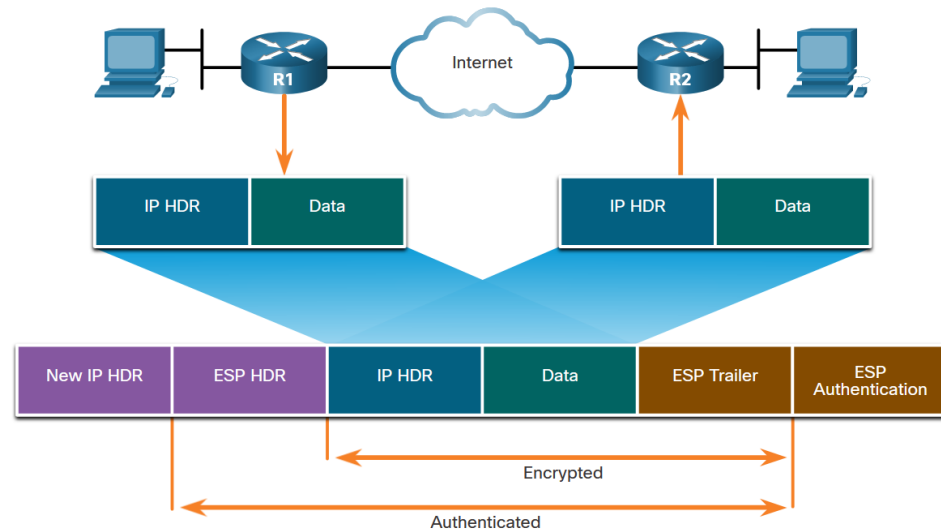
Optionally, ESP can also enforce anti-replay protection which verifies that each packet is unique and is not duplicated. This protection ensures that a hacker cannot intercept packets and insert changed packets into the data stream. Anti-replay works by keeping track of packet sequence numbers and using a sliding window on the destination end.

When a connection is established between a source and destination, their counters are initialized at zero. Each time a packet is sent, a sequence number is appended to the packet by the source. The destination uses the sliding window to determine which sequence numbers are expected. The destination verifies that the sequence number of the packet is not duplicated and is received in the correct order.

ESP Encrypts and Authenticates

When both authentication and encryption are selected, encryption is performed first. One reason for this order of processing is that it facilitates rapid detection and rejection of replayed or bogus packets by the receiving device. Prior to decrypting the packet, the receiver can authenticate inbound packets.

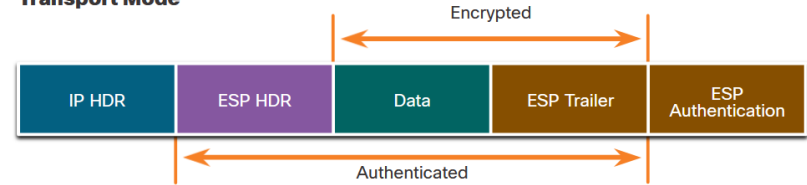
IPsec was initially established to provide security for IPv6 packets. Therefore, the IPsec implementations for IPv4 and IPv6 are similar as far as the standards are concerned. In IPv4, AH and ESP are IP protocol headers. IPv6 uses the extension headers with a next-header value of 50 for ESP and 51 for AH.



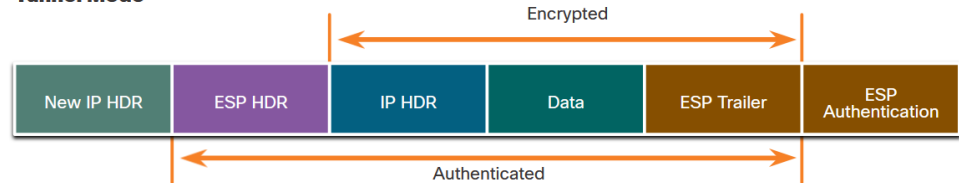
Transport and Tunnel Modes

- Transport Mode** - In transport mode, security is provided only for the transport layer of the OSI model and above. Transport mode protects the payload of the packet but leaves the original IP address in plaintext. The original IP address is used to route the packet through the internet. ESP transport mode is used between hosts.
- Tunnel Mode** - Tunnel mode provides security for the complete original IP packet. The original IP packet is encrypted and then it is encapsulated in another IP packet. This is known as IP-in-IP encryption. The IP address on the outside IP packet is used to route the packet through the internet.

Transport Mode



Tunnel Mode



18.5 Internet Key Exchange

Internet Key Exchange

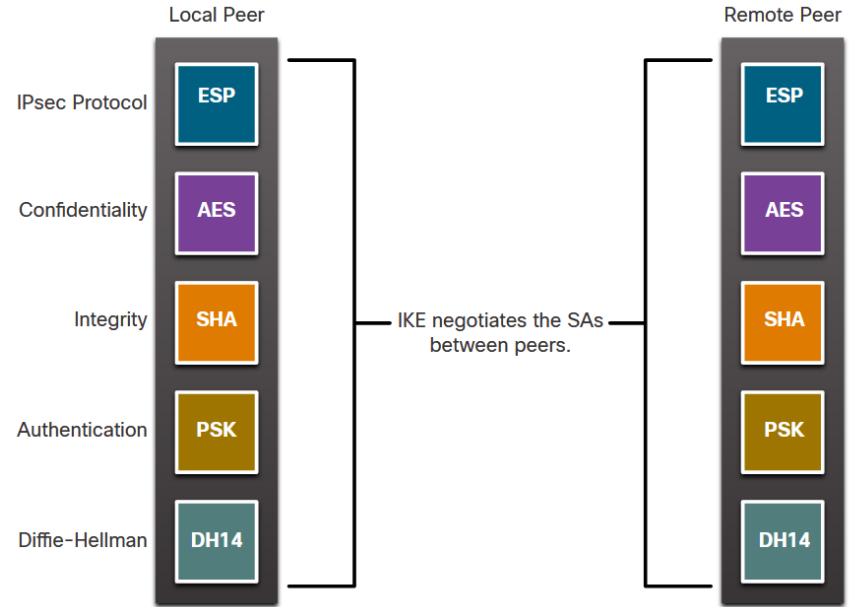
The IKE Protocol

IKE enhances IPsec by adding features and simplifies configuration for the IPsec standard.

IKE is a hybrid protocol that implements key exchange protocols inside the Internet Security Association Key Management Protocol (ISAKMP) framework.

Instead of transmitting keys directly across a network, IKE calculates shared keys based on the exchange of a series of data packets.

UDP port 500 packets must be permitted on any IP interface that is connecting a security gateway peer.



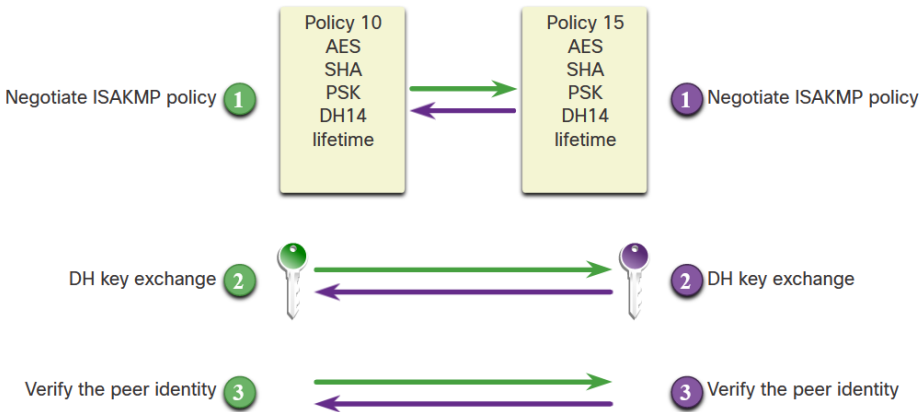
Internet Key Exchange

Phase 1 and 2 Key Negotiation

IKE uses ISAKMP for phase 1 and phase 2 of key negotiation. Phase 1 negotiates a security association (a key) between two IKE peers. The key negotiated in phase 1 enables IKE peers to communicate securely in phase 2. During phase 2 negotiation, IKE establishes keys (security associations) for other applications, such as IPsec.

In Phase 1, two IPsec peers perform the initial negotiation of SAs. The basic purpose of Phase 1 is to negotiate ISAKMP policy, authenticate the peers, and set up a secure tunnel between the peers. This tunnel will then be used in Phase 2 to negotiate the IPsec policy.

Phase 1 – Negotiate ISAKMP policy to create a tunnel.

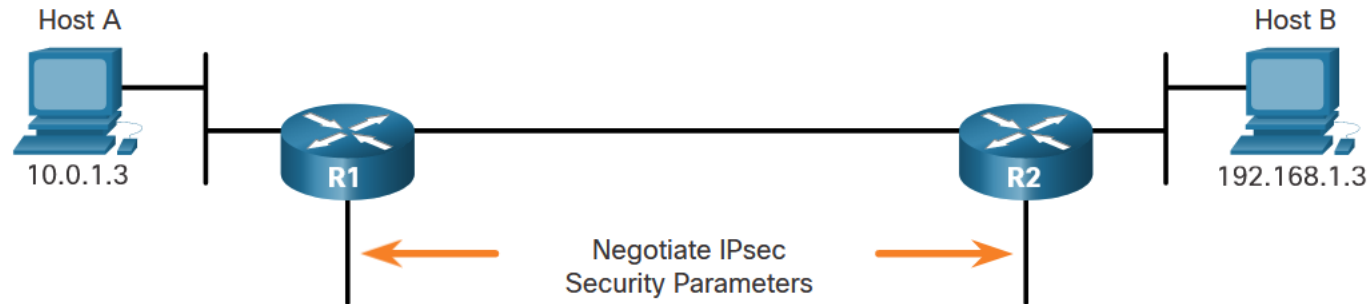


Phase 2 – Negotiate IPsec policy for sending secure traffic across the tunnel.



Phase 2: Negotiating SAs

IKE Phase 2 is called quick mode and can only occur after IKE has established a secure tunnel in Phase 1. Quick mode negotiates the IKE Phase 2 SAs. In this phase, the SAs that IPsec uses are unidirectional; therefore, a separate key exchange is required for each data flow.



Video - IKE Phase 1 and Phase 2

This video will explain IPsec: Confidentiality, Integrity, Authentication, and Secure Key Exchange.

18.6 VPNs Summary

What Did I Learn in this Module?

- Organizations use virtual private networks (VPNs) to create end-to-end private network connections that are transported over a public network.
- VPNs support encryption features, such as IPsec and SSL to secure network traffic between sites.
- Site-to-site VPNs are created when VPN gateways are preconfigured with information to establish a secure tunnel.
- Remote access VPNs (clientless and client-based) are used to securely connect remote and mobile users to the enterprise by creating a secure tunnel.
- IPsec is a framework used to define how a VPN connection will ensure confidentiality, integrity, and origin authentication.
- When establishing a VPN link, the peers must share the same SA to negotiate key exchange parameters, establish a shared key, authenticate each other, and negotiate the encryption parameters.
- The two main IPsec protocols are AH and ESP, which can be applied to IP packets using transport mode or tunnel mode.

What Did I Learn in this Module? (Cont.)

- The IKE protocol is a key management protocol standard that is used to automatically negotiate IPsec security associations and enable IPsec secure communications.
- IKE uses ISAKMP for phase 1 and phase 2 of key negotiation. Phase 1 negotiates a security association (a key) between two IKE peers. During phase 2 negotiation, IKE establishes keys (security associations) for other applications, such as IPsec.

New Terms and Commands

- virtual private networks (VPNs)
- tunnel mode
- Authentication Header (AH)
- remote-access VPN
- Internet Protocol Security (IPsec)
- Internet Security Association Key Management Protocol (ISAKMP)
- Encapsulation Security Protocol (ESP)
- security association (SA)
- site-to-site VPN
- transport mode
- pre-shared secret key (PSK)

